Exp 1 · Zener diode characteristics and dynamic resistance

Manual pp. 7–8 · hands-on · CO1, CO2 · theory: ECE2101 unit 2 (pn junctions)

0 Instruments → **1 Zener curve** → 2 Regulator → 3 Clipper → 4 Clamper

What you already have: Ohm's law, and the meters from Exp 0.

Three terms first:

- **Diode:** a p-type and an n-type piece of silicon joined together (a *pn junction*). The p side is the **anode** and the n side is the **cathode** (the bar on the symbol). Current flows easily from anode to cathode.
- **Forward / reverse bias:** anode more positive than cathode = forward. The reverse = reverse.
- **Breakdown:** under enough reverse voltage, a junction suddenly conducts heavily.

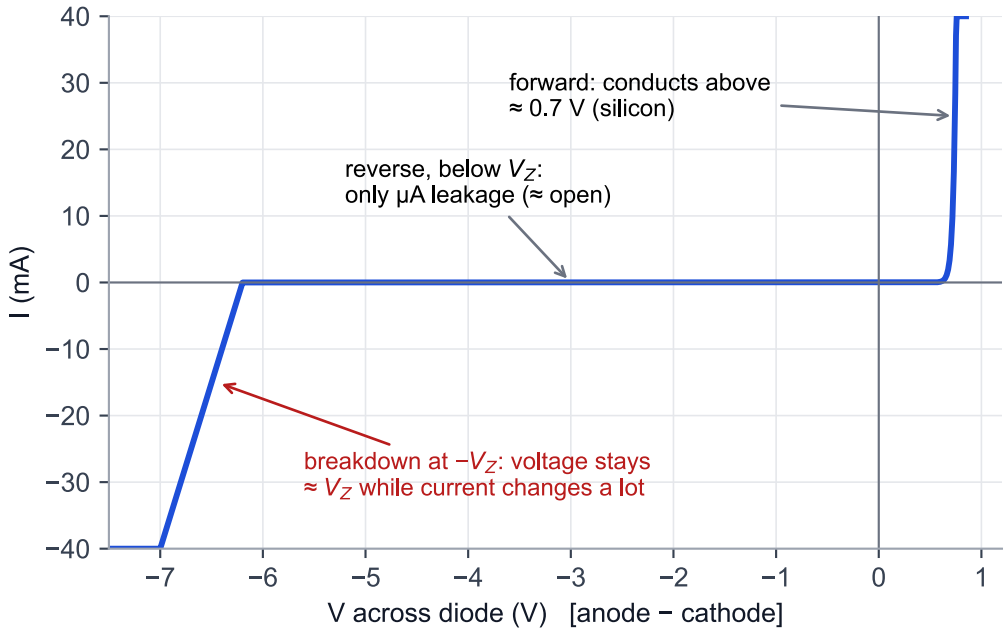
MANUAL EXP 1, AIM

To plot the characteristics of a zener diode and find its dynamic resistance under reverse-biased condition. Viva form: "Why does a zener's reverse current rise sharply at one voltage, and what is its *dynamic* resistance?"

GUESS FIRST

Sketch I against V for an ordinary diode, both directions, before reading on. Where does it conduct?

The idea: a diode that is meant to break down



One curve, three regions. A zener is built so the left-hand breakdown happens at a chosen, sharp voltage V_Z.

1. **Forward:** almost no current until about 0.6–0.7 V (silicon), then current rises steeply. This is the same as any diode.
2. **Reverse, below V_Z:** only a tiny leakage current (μA). The diode is close to an open circuit.
3. **Reverse, at V_Z:** the junction **breaks down**. The current can rise from 1 mA to 10 mA while the voltage barely moves from V_Z.

Region 3 is what makes a zener useful: **a voltage that stays put while the current changes**, which is a voltage reference. Exp 2 uses exactly that.

Why breakdown happens (viva staple; from ECE2101 unit 2):

	Zener breakdown	Avalanche breakdown
doping	heavy, so the junction is thin	light, so the junction is wide
mechanism	the field pulls electrons straight across (tunnelling)	fast carriers knock out new ones (chain reaction)
V_Z	below about 5 V	above about 6 V
temperature coefficient	negative	positive

A 6.2 V zener is near the crossover and uses both mechanisms. That is why ~5–6 V zeners drift least with temperature. "Zener diode" is the trade name for any breakdown diode, whichever mechanism it uses.

Dynamic resistance r_z: in breakdown, how much the voltage moves per unit change of current:

$$r_z = \frac{\Delta V_Z}{\Delta I_Z}$$

A small r_z (tens of ohms or less) means a better voltage reference. It is called *dynamic* because it is the slope of the curve, not V/I at one point. V_Z/I_Z would be a static resistance, and a meaningless one here.

Limits:

- **$I_{Z(min)}$:** the least current that keeps the zener in breakdown.
- **$I_{Z(max)} = P_Z/V_Z$:** the largest current before it overheats. A 0.5 W, 6.2 V zener allows $I_{Z(max)} = 0.5/6.2 = 81$ mA.

The series 1 kΩ resistor in the circuit limits the current.

The circuit

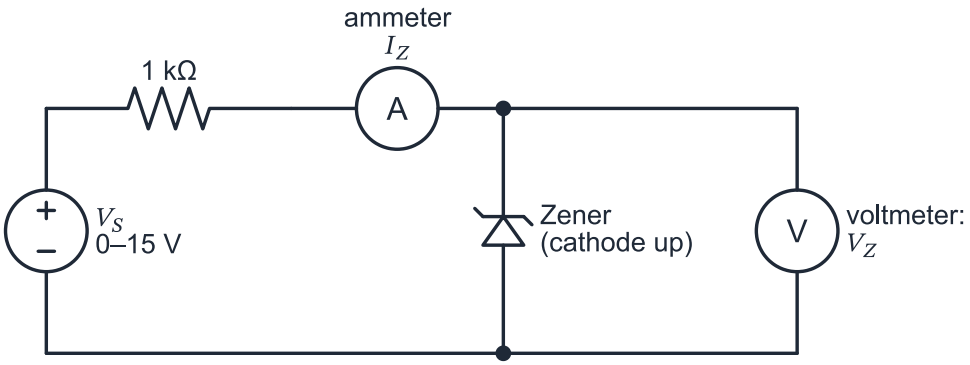


Fig 1.1, redrawn. The zener's cathode faces the + side, so it is reverse-biased. The ammeter is in series; the voltmeter (or DSO) is across the zener.

TRAP: THE MANUAL'S PARTS LIST

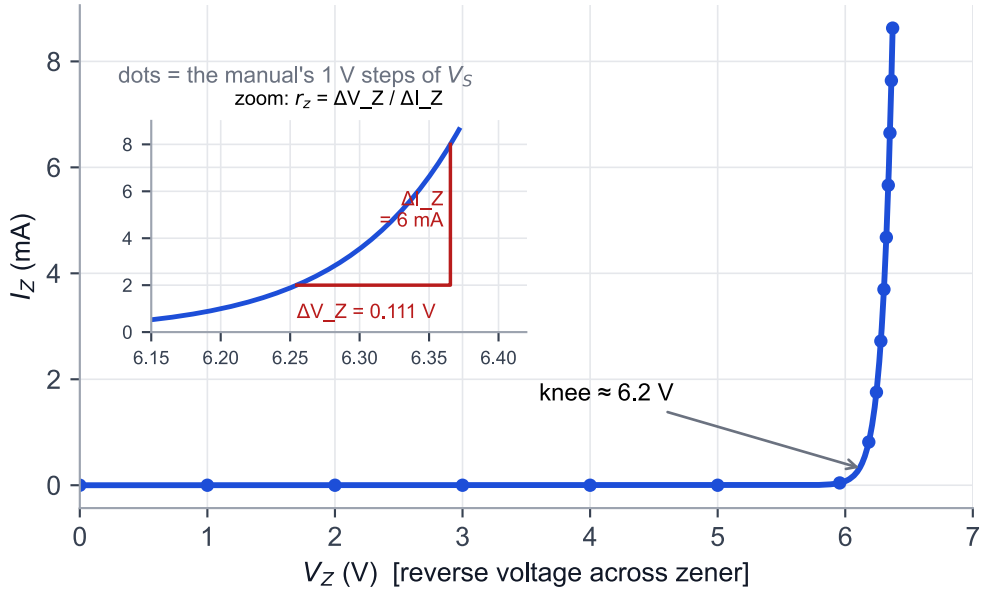
It lists a **1N4007**. That is an ordinary rectifier with a breakdown near 1000 V, so it will never break down on a 0–30 V supply. Ask for a **zener** (e.g. 6.2 V). A 1N4007 placed reverse-biased just shows µA leakage at every V_S.

DO IT (HANDS-ON, BREADBOARD)

1. Supply **off**, voltage knob at 0, current limit low.
2. Wire the 1 kΩ from supply + to the ammeter. Ammeter (DMM on mA DC, red lead in the mA port) to the zener's **cathode** (the banded end). Zener's anode to supply −.
3. Put the voltmeter (DMM on DC V, or the DSO) across the zener.
4. Switch on. Raise V_S in **1 V steps from 0 to 15 V**, reading V_Z and I_Z at each step.
5. Plot I_Z (y) against V_Z (x). Mark the knee: that is V_Z.
6. Take two points well into breakdown, read ΔV_Z and ΔI_Z, and calculate r_z.

What you should see

Simulated with a 6.2 V zener (BZX84C6V2L, the part Exp 2 uses). A different zener shifts the knee to its own V_Z, but the shape is identical.



LTspice, 1 kΩ series, V_S 0–15 V. Flat (open) until about 6 V, then nearly vertical. The zoom shows the r_z triangle.

V_S (V)	0–5	6	7	8	9	10	11	12	13	14	15
V_Z (V)	= V_S	5.957	6.185	6.245	6.279	6.303	6.322	6.337	6.350	6.362	6.372
I_Z (mA)	≈ 0 (µA)	0.04	0.82	1.76	2.72	3.70	4.68	5.66	6.65	7.64	8.63

Below breakdown, the zener takes almost no current, so no voltage drops across the 1 kΩ and $V_Z = V_S$. In breakdown, V_Z is pinned and the rest of V_S drops across the resistor: $I_Z = (V_S - V_Z)/1\ k\Omega$. Check: $V_S = 15$ gives $(15 - 6.372)/1000 = 8.63$ mA ✓.

Calculation

Using the 10 V and 15 V rows (both well into breakdown):

$$r_z = \frac{6.372 - 6.303}{8.63 - 3.70\ \text{mA}} = \frac{0.069\ \text{V}}{4.93\ \text{mA}} \approx 14\ \Omega$$

The zoom figure uses 2 → 8 mA and gives 0.111 V / 6 mA = 18 Ω. r_z shrinks as the current grows, so always **state the current range used**.

WRITE IN YOUR RECORD

Result: breakdown voltage $V_Z \approx$ **6.2 V** (at $I_Z \approx 1$ mA); dynamic resistance $r_z \approx$ **14 Ω** (between 3.7 and 8.6 mA). Use your own readings in the lab.

Theory for the record: a zener is a heavily doped pn diode designed to operate in reverse breakdown. Forward, it behaves like a normal diode (about 0.7 V). Reverse, only leakage flows until V_Z, where the current rises sharply while the voltage stays nearly constant. Breakdown is by tunnelling (below ~5 V) or avalanche (above ~6 V). Operation must stay between $I_{Z(min)}$ and $I_{Z(max)} = P_Z/V_Z$. The slope $\Delta V_Z/\Delta I_Z$ in breakdown is the dynamic resistance; a small value means good regulation, which Exp 2 uses.

TRAP: NO SERIES RESISTOR

Without the 1 kΩ (or with it shorted), the current in breakdown is limited only by the supply, and the zener burns out. Never connect a zener directly across a supply.

CHECK 1

(a) At $V_S = 12$ V, a student reads $V_Z = 6.34$ V. Predict I_Z without the ammeter. (b) Why is r_z not simply V_Z/I_Z? (c) A 3.3 V zener and a 12 V zener: which mechanism does each use, and which way does each one's V_Z move when heated?

Answers

Opening question. Reverse current rises sharply at V_Z because the junction breaks down (tunnelling or avalanche), producing many carriers. Dynamic resistance = $\Delta V_Z/\Delta I_Z$, the slope in breakdown. It is small (≈ 14–18 Ω here).

Check 1. (a) I_Z = (12 − 6.34)/1 kΩ = **5.66 mA**. (b) r_z describes how V changes when I changes (the slope), which is what matters for regulation. V_Z/I_Z at one point is a large, current-dependent number with no use here. (c) 3.3 V: Zener (tunnelling) mechanism, V_Z falls with heat (negative coefficient). 12 V: avalanche, V_Z rises with heat (positive coefficient).